Assignment-1

Q1/Solution:

(a) $5\bar{A}-2\bar{B} = 5(-4\hat{x}+2\hat{y}+3\hat{z})-2(3\hat{x}+4\hat{y}-\hat{z})$

$= -26\hat{x}+2\hat{y}+17\hat{z}$

$|5\bar{A}-2\bar{B}| = \sqrt{(-26)^2+(2)^2+(17)^2} = 31.129$ ANS

(b) $\dfrac{5\bar{A}-2\bar{B}}{|\bar{A}|} = \dfrac{-26\hat{x}+2\hat{y}+17\hat{z}}{\sqrt{(-4)^2+(2)^2+(3)^2}} = \bar{G}$ (Let)

$\bar{G} = -4.828\hat{x}+0.371\hat{y}+3.157\hat{z}$

Hence unit vector in the direction of $\bar{G}$

$\hat{G} = \dfrac{-4.828\hat{x}+0.371\hat{y}+3.157\hat{z}}{\sqrt{(-4.828)^2+(0.371)^2+(3.157)^2}}$

$\hat{G} = -0.835\hat{x}+0.064\hat{y}+0.546\hat{z}$ ← ANS

(c): comp. of $\bar{A}$ // to $\bar{B}$

$= (\bar{A}\cdot\hat{B})\hat{B} = \left(\bar{A}\cdot\dfrac{\bar{B}}{|\bar{B}|}\right)\dfrac{\bar{B}}{|\bar{B}|}$

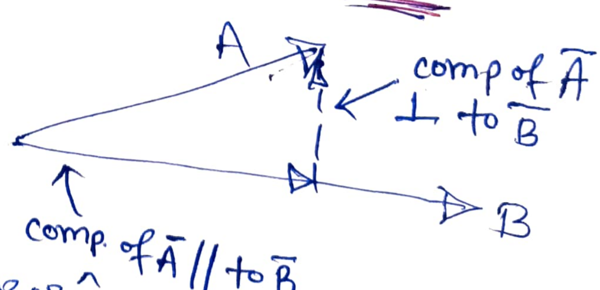
$= \left[\dfrac{(-4)(3)+(2)(4)+3(-1)}{\{(3)^2+(4)^2+(-1)^2\}^{1/2}}\right]\left[\dfrac{3\hat{x}+4\hat{y}-\hat{z}}{\sqrt{3^2+4^2+(-1)^2}}\right]$

$= -0.808\hat{x}-1.077\hat{y}+0.269\hat{z}$ ← ANS

(d)

∴ comp. of $\bar{A} \perp \bar{B}$

$= \bar{A} - \text{comp } \bar{A} // \bar{B}$



$= (-4\hat{x}+2\hat{y}+3\hat{z})-(-0.808\hat{x}-1.077\hat{y}+0.269\hat{z})$

$= -3.192\hat{x}+3.077\hat{y}+2.731\hat{z}$ ← ANS.

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Q2/ solution: Cartesign coordinates of B

(a) $x = 4\cos(-50^\circ) = 2.571$, $y = 4\sin(-50^\circ) = -3.064$

$z = 2 \rightarrow B(2.571, -3.064, 2)$

$R_{BA} = \bar{r}_A - \bar{r}_B = (2\hat{x} + 3\hat{y} - \hat{z}) - (2.571\hat{x} - 3.064\hat{y} + 2\hat{z})$

$= -0.571\hat{x} + 6.064\hat{y} - 3\hat{z} = \bar{A}$ (let)

Let $R_{BA} = A_\rho\hat{\rho} + A_\phi\hat{\phi} + A_z\hat{z} =$

$A_\rho = \bar{A}\cdot\rho = -0.571\hat{x}\cdot\hat{\rho} + 6.064\hat{y}\cdot\hat{\rho} - 3\hat{z}\cdot\hat{\rho}$

$= -0.571\cos\phi + 6.064\sin\phi$

$A_\phi = \bar{A}\cdot\hat{\phi} = -0.571\hat{x}\cdot\hat{\phi} + 6.064\hat{y}\cdot\hat{\phi} - 3\hat{z}\cdot\hat{\phi}$

$= 0.571\sin\phi + 6.064\cos\phi$

$A_z = \bar{A}\cdot\hat{z} = (-0.571\hat{x}\cdot\hat{z} + 6.064\hat{y}\cdot\hat{z} - 3\hat{z}\cdot\hat{z} = +3$

$\bar{A} = \bar{R}_{BA} = (-0.571\cos\phi + 6.064\sin\phi)\hat{\rho} + (0.571\sin\phi + 6.064\cos\phi)\hat{\phi} - 3\hat{z}$

Subs. ϕ coordinate at B into $\bar{A}(=R_{BA})$

$\bar{R}_{BA}$ at $B = -5.012\hat{\rho} + 3.460\hat{\phi} - 3\hat{z}$

$$\bar{a}_{BA} = \frac{\bar{R}_{BA}}{|R_{BA}|} = \frac{-5.012\hat{\rho} + 3.460\hat{\phi} - 3\hat{z}}{\sqrt{(-5.012)^2 + (3.460)^2 + (-3)^2}}$$

$= -0.738\hat{\rho} + 0.510\hat{\phi} - 0.442\hat{z}$ ANS

(b) $\bar{R}_{AB} = \bar{r}_B - \bar{r}_A = (2.571\hat{x} - 3.064\hat{y} + 2\hat{z}) - (2\hat{x} + 3\hat{y} - \hat{z})$

$= (0.571\hat{x} - 6.064\hat{y} + 3\hat{z})$

Let $\bar{B} = \bar{R}_{AB} = B_\rho\hat{\rho} + B_\phi\hat{\phi} + B_z\hat{z}$

$B_\rho = \bar{B}\cdot\hat{\rho} = 0.571\hat{x}\cdot\hat{\rho} - 6.064\hat{y}\cdot\hat{\rho} + 3\hat{z}\cdot\hat{\rho}$

$= 0.571\cos\phi - 6.064\sin\phi$

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Q2(b)... contd.

$$B_\phi = \bar{B}\cdot\hat{\phi} = 0.571\,\hat{x}\cdot\hat{\phi} - 6.064\,\hat{y}\cdot\hat{\phi} + 3\,\hat{z}\cdot\hat{\phi}$$

$$= -0.571\sin\phi - 6.064\cos\phi$$

$$B_z = \bar{B}\cdot\hat{z} = 0.571\,\hat{x}\cdot\hat{z} - 6.064\,\hat{y}\cdot\hat{z} + 3\,\hat{z}\cdot\hat{z}$$

$$= 3$$

$$\therefore \bar{B} = \bar{R}_{AB} = (0.571\cos\phi - 6.064\sin\phi)\,\hat{\rho} + (-0.571\sin\phi - 6.064\cos\phi)\,\hat{\phi} + 3\hat{z}$$

For Pt. A, $\phi = \tan^{-1}(3/2) = 56.31^\circ$

Put ϕ at A into $\bar{B}$

So Vector at Pt. $\bar{A}$ $= -4.729\,\hat{\rho} - 3.839\,\hat{\phi} + 3\hat{z}$

$$\hat{R}_{AB} = \frac{\bar{R}_{AB}}{|\bar{R}_{AB}|} = \frac{-4.729\,\hat{\rho} - 3.829\,\hat{\phi} + 3\hat{z}}{\sqrt{(-4.729)^2 + (-3.829)^2 + (3)^2}}$$

$$= \underline{-0.696\,\hat{\rho} - 0.565\,\hat{\phi} + .442\,\hat{z}}$$

ANS

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Sol:

Q3:(a) $\bar{R}_{AB} = \bar{r}_B - \bar{r}_A = (\hat{x} + 3\hat{y} + 4\hat{z}) - (2\hat{x} - \hat{y} - 3\hat{z})$

$= -\hat{x} + 4\hat{y} + 7\hat{z}$ $\longleftarrow$ ANS.

(b) Let $\bar{A} = -\hat{x} + 4\hat{y} + 7\hat{z} = A_\rho \hat{\rho} + A_\phi \hat{\phi} + A_z \hat{z}$

$A_\rho = \bar{A}\cdot\hat{\rho} = -\hat{x}\cdot\hat{\rho} + 4\hat{y}\cdot\hat{\rho} + 7\hat{z}\cdot\hat{\rho} = -\cos\phi + 4\sin\phi$

$A_\phi = \bar{A}\cdot\hat{\phi} = -\hat{x}\cdot\hat{\phi} + 4\hat{y}\cdot\hat{\phi} + 7\hat{z}\cdot\hat{\phi} = \sin\phi + 4\cos\phi$

$A_z = \bar{A}\cdot\hat{z} = -\hat{x}\cdot\hat{z} + 4\hat{y}\cdot\hat{z} + 7\hat{z}\cdot\hat{z} = 7$

$\bar{A} = \bar{R}_{AB} = (-\cos\phi + 4\sin\phi)\hat{\rho} + (\sin\phi + 4\cos\phi)\hat{\phi} + 7\hat{z}$

For point A $\phi = \tan^{-1}\left(-\frac{1}{2}\right) = -26.565°$

put value of ϕ into $\bar{R}_{AB}$

$\therefore$ $\bar{R}_{AB}$ at ~~Pt.~~ A $= -2.683\hat{\rho} + 3.130\hat{\phi} + 7\hat{z}$

ANS.

(c) Let $\bar{A} = A_r\hat{r} + A_\theta\hat{\theta} + A_\phi\hat{\phi}$

$A_r = \bar{A}\cdot\hat{r} = -\hat{x}\cdot\hat{r} + 4\hat{y}\cdot\hat{r} + 7\hat{z}\cdot\hat{r}$

$= -\sin\theta\cos\phi + 4\sin\theta\sin\phi + 7\cos\theta$

$A_\theta = \bar{A}\cdot\hat{\theta} = -\hat{x}\cdot\hat{\theta} + 4\hat{y}\cdot\hat{\theta} + 7\hat{z}\cdot\hat{\theta}$

$= -\cos\theta\cos\phi + 4\cos\theta\sin\phi - 7\sin\theta$

$A_\phi = \bar{A}\cdot\hat{\phi} = -\hat{x}\cdot\hat{\phi} + 4\hat{y}\cdot\hat{\phi} + 7\hat{z}\cdot\hat{\phi} = \sin\phi + 4\cos\phi$

$\bar{R}_{AB} = (-\sin\theta\cos\phi + 4\sin\theta\sin\phi + 7\cos\theta)\hat{r} + (-\cos\theta\cos\phi + 4\cos\theta\sin\phi - 7\sin\theta)\hat{\theta} + (\sin\phi + 4\cos\phi)\hat{\phi}$

at A, ~~r~~ $\theta = \cos^{-1}(-3/3.742) = 143.29°$, $\phi = \tan^{-1}(-1/2) = -26.565°$

$r = \sqrt{2^2 + (-1)^2 + (-3)^2} = 3.742$

Put values of θ & ϕ into $\bar{R}_{AB}$

$\bar{R}_{AB} = -7.216\hat{r} - 2.033\hat{\theta} + 3.130\hat{\phi}$ $\Leftarrow$ ANS

Assignment - 1

Q4/solution :

(a) $$V = \iiint dv = \int_3^5 \int_{0.1\pi}^{0.3\pi} \int_{1.2\pi}^{1.6\pi} r^2 \sin\theta \, dr\, d\theta\, d\phi$$

$$= \left(\frac{r^3}{3}\right)_3^5 \cdot \left[-\cos\theta\right]_{0.1\pi}^{0.3\pi} \cdot \left[\phi\right]_{1.2\pi}^{1.6\pi} = \underline{14.912} \leftarrow \text{Ans}$$

(b) $$d(P_1P_2) = \left[\left(3\sin(0.1\pi)\cdot\cos(1.2\pi) - 5\sin(0.3\pi)\cos(1.6\pi)\right)^2 + \left(3\sin(0.1\pi)\cdot\sin(1.2\pi) - 5\sin(0.3\pi)\cdot\sin(1.6\pi)\right)^2 + \left(3\cos(0.1\pi) - 5\cos(0.3\pi)\right)^2 \right]^{1/2}$$

$$= \underline{3.862} \longleftarrow \text{ANS}$$

(c) Surface $S = S_1 + S_2 + S_3 + S_4 + S_5 + S_6$

$$S = \int_{.1\pi}^{.3\pi} \int_{1.2\pi}^{1.6\pi} r^2 \sin\theta \, d\theta\, d\phi \Big|_{r=3} + \int_{0.1\pi}^{0.3\pi} \int_{1.2\pi}^{1.6\pi} r^2 \sin\theta \, d\theta\, d\phi \Big|_{r=5} + \int_3^5 \int_{1.2\pi}^{1.6\pi} r\sin\theta \, dr\, d\phi \Big|_{\theta=0.1\pi} + \int_3^5 \int_{1.2\pi}^{1.6\pi} r\sin\theta\, dr\, d\phi \Big|_{\theta=.3\pi} + \int_3^5 \int_{.1\pi}^{.3\pi} r\, dr\, d\theta \Big|_{\phi=1.2\pi} + \int_3^5 \int_{.1\pi}^{.3\pi} r\, dr\, d\phi \Big|_{\phi=1.6\pi}$$

$$S = (3)^2 \left[-\cos\theta\right]_{.1\pi}^{.3\pi} \left[\phi\right]_{1.2\pi}^{1.6\pi} + (5)^2 \left[-\cos\theta\right]_{.1\pi}^{.3\pi} \left[\phi\right]_{1.2\pi}^{1.6\pi} + \sin(0.1\pi) \left[\frac{r^2}{2}\right]_3^5 \left[\phi\right]_{1.2\pi}^{1.6\pi} + \sin(0.3\pi) \left[\frac{r^2}{2}\right]_3^5 \left[\phi\right]_{1.2\pi}^{1.6\pi} + \left[\frac{r^2}{2}\right]_3^5 \left[\theta\right]_{0.1\pi}^{0.3\pi} + \left[\frac{r^2}{2}\right]_2^5 \cdot \left[\theta\right]_{.1\pi}^{.3\pi} = \underline{36.814} \text{ Ans.}$$

Assigment - 1

Q5/solution

(a) $\bar{E} = \frac{100\cos 60^\circ}{(2)^3}\hat{r} + \frac{50\sin 60^\circ}{(2)^3}\hat{\theta}$

$= 6.25\hat{r} + 5.413\hat{\theta}$

$|\bar{E}| = \sqrt{(6.25)^2 + (5.413)^2} = 8.268$

(b) $\bar{E} = E_x\hat{x} + E_y\hat{y} + E_z\hat{z}$

$E_x = \bar{E}\cdot\hat{x} = 6.25\hat{r}\cdot\hat{x} + 5.413\hat{\theta}\cdot\hat{x}$

$= 6.25\sin\theta\cos\phi + 5.413\cos\theta\cos\phi$

$E_y = \bar{E}\cdot\hat{y} = 6.25\hat{r}\cdot\hat{y} + 5.413\hat{\theta}\cdot\hat{y}$

$= 6.25\sin\theta\sin\phi + 5.413\cos\theta\sin\phi$

$E_z = \bar{E}\cdot\hat{z} = 6.25\hat{r}\cdot\hat{z} + 5.413\hat{\theta}\cdot\hat{z}$

$= 6.25\cos\theta - 5.413\sin\theta$

$\therefore \bar{E} = (6.25\sin\theta\cos\phi + 5.413\cos\theta\cos\phi)\hat{x} +$
$(6.25\sin\theta\sin\phi + 5.413\cos\theta\sin\phi)\hat{y} +$
$(6.25\cos\theta - 5.413\sin\theta)\hat{z}$

Subst. $\theta = 60$, & $\phi = 20^\circ$ into $\bar{E}$

E at Pt. $(2, 60^\circ, 20^\circ) = 7.630\hat{x} + 2.777\hat{y} - 1.563\hat{z}$

$\hat{E} = \frac{\bar{E}}{|\bar{E}|} = \frac{7.630\hat{x} + 2.777\hat{y} - 1.563\hat{z}}{\sqrt{(7.630)^2 + (2.777)^2 + (-1.563)^2}}$

$= 0.923\hat{x} + 0.336\hat{y} - 0.189\hat{z}$ ← ANS

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